Executive Summary

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Motivation and Research Question

High and volatile inflation has historically imposed substantial economic costs, particularly in countries where monetary policy is subject to political pressure. One widely discussed institutional solution is greater central bank independence, which is intended to insulate monetary policy from short term political incentives. While theory predicts that more independent central banks should deliver lower inflation, identifying the timing and magnitude of this relationship in the data remains challenging.

This project asks the following question:
Are increases in central banking independence associated with lower future inflation?

Rather than comparing countries with different average levels of independence, we focus on institutional reforms within countries and examine how inflation evolves before and after those reforms occur.

Measuring Central Bank Independence

We measure central bank independence using the Central Bank Independence Extended index developed by Romelli. This index is a de jure measure based on central bank statutes and ranges from zero to one, with higher values indicating greater independence. It aggregates forty two legal provisions across six dimensions, including the appointment and dismissal of central bank leadership, control over monetary policy, limits on lending to the government, financial independence, and reporting and disclosure requirements.

Because the index is based on legal design rather than observed policy outcomes, it captures institutional intent rather than enforcement. However, some components of the index may be correlated with broader political stability or governance quality. This motivates our use of fixed effects and careful interpretation of results.

Data and Reform Identification

We combine the central bank independence data with annual inflation data from the International Monetary Fund. To improve comparability across countries, we restrict attention to reforms occurring after 1980, which we treat as the modern era of central banking.

For each country, we compute year to year changes in the independence index and define a reform as the **first positive change** observed after 1980. This approach isolates the initial institutional shift toward greater independence and avoids conflating multiple incremental reforms. Each country is assigned a single reform year, which is then merged into the inflation panel.

We construct an event time variable that measures the number of years relative to a country's reform. Negative values indicate years before the reform, while positive values indicate years after. To ensure consistent comparisons across countries, we restrict the sample to a symmetric window around the reform, typically ten years before and ten years after.

Empirical Approach

Our main analysis uses an event study framework. Inflation is regressed on a full set of event time indicator variables, which capture how inflation evolves in each year relative to the reform. One relative year is omitted as the reference category.

We include country fixed effects to control for all time invariant differences between countries, such as geography, historical institutions, or long run economic development. We also include decade fixed effects to capture global inflation regimes and worldwide macroeconomic shocks that affect many countries simultaneously. Because reform timing is highly clustered across countries, calendar-year fixed effects are collinear with event-time indicators; we therefore include decade fixed effects to capture global inflation regimes instead of year fixed effects.

Standard errors are clustered by country to account for persistent within country inflation dynamics over time. This ensures that statistical inference is not driven by serial correlation.

This approach allows us to use the full panel structure of the data and trace the dynamic relationship between institutional reform and inflation, rather than relying on simple before and after comparisons.

Main Findings

The event study results show no strong evidence of differential inflation trends prior to reform, which supports the parallel trends assumption underlying the analysis. In the full sample, inflation declines gradually following reform, consistent with crisis-era stabilization dynamics (Figure 1). However, when reforms occurring during high-inflation episodes are excluded and likely crisis reforms are dropped from treatment, inflation dynamics are flat both before and after reform (Figure 2). This suggests that the observed disinflation in the full sample is driven primarily by crisis-driven reforms rather than by institutional reform per se.

These patterns are consistent with the idea that institutional reforms affect inflation through credibility and expectations rather than abrupt policy changes.

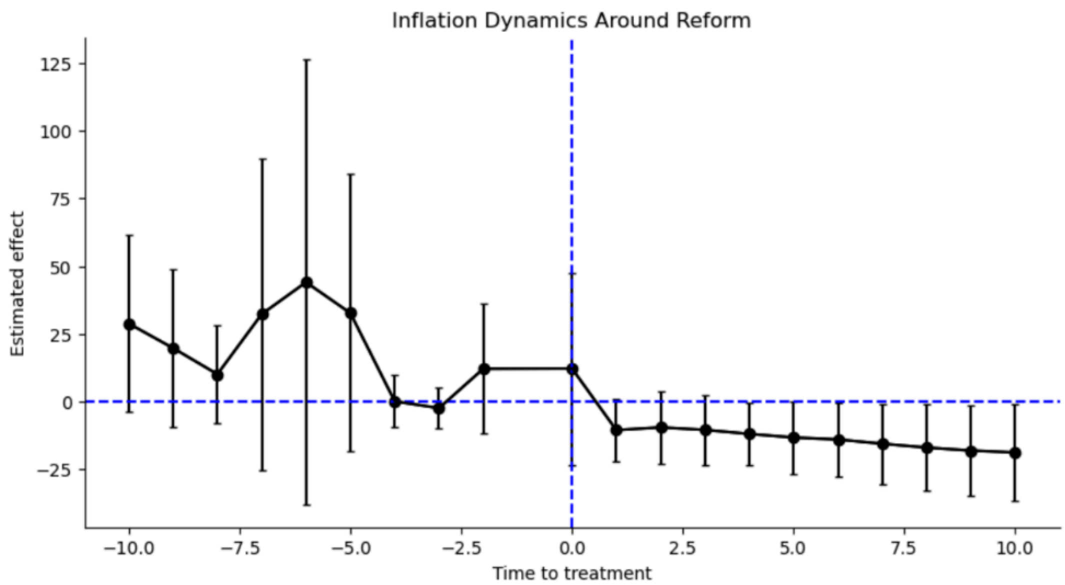

Figure 1

Note: The x axis shows years relative to the reform year. The y axis shows estimated effect size on inflation. This figure shows how inflation evolves over time around the reform year, providing a descriptive view of inflation dynamics before and after increases in central bank independence. Also note the Y axis values being very large in magnitude showing a negative effect size post treatment.

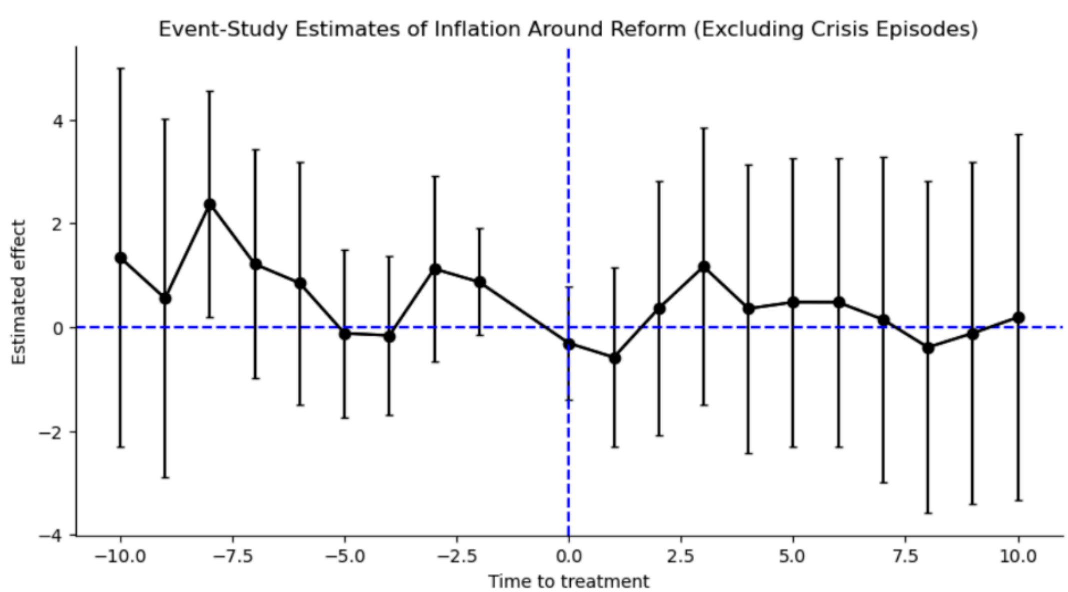

Figure 2

Note: The x axis shows years relative to the reform year. The y axis shows estimated effect size on inflation. The absence of strong pre reform trends supports the parallel trends assumption, and negative post reform estimates indicate declining inflation following reform. Crisis episodes are excluded from this specification. Based on post-treatment effects, this more robust model appears to have much more null effects to treatment.

Robustness and Limitations

Several robustness checks were conducted. First, we test for pre reform trends using a joint hypothesis test of all pre reform event coefficients. The test fails to reject the null that these coefficients are jointly zero, providing evidence in favor of parallel trends. Second, we exclude reforms that occur during or immediately around extreme inflation episodes to address concerns about crisis driven reforms. The qualitative pattern differs across samples, highlighting the role of crisis-driven reforms in shaping inflation dynamics

Despite these checks, important limitations remain. Reverse causality is a concern if declining inflation itself makes reforms politically feasible. Omitted variables related to political or economic stability may still bias estimates, even with fixed effects. Measurement error is also possible, as legal independence does not always translate into practical independence.

For these reasons, our findings should be interpreted as **conditional correlations** rather than definitive causal effects.

Conclusion

This project finds that inflation declines following reforms that occur during high-inflation episodes, while reforms undertaken outside of crises are largely inflation-neutral. These results suggest that the relationship between central bank independence and inflation is mediated by macroeconomic conditions at the time of reform. Using an event study framework allows us to trace these dynamics over time while controlling for country specific factors and global inflation trends.

Real world episodes such as Venezuela in the 2000s illustrate how sharp reductions in central bank independence (through corruption etc.) can enable sustained monetary financing of government deficits, contributing to persistent increases in inflation, a pattern that mirrors the dynamic relationship observed in our analysis in the opposite direction.

While the results are consistent with theoretical predictions, they should be interpreted cautiously. Nonetheless, the analysis provides evidence that institutional design may play an important role in shaping macroeconomic outcomes over the medium to long run.

Specification	T = 0	T = +5	T = +10
(1) Full sample	12.1835	-13.2895*	-18.8388**
(2) Excluding crisis reforms and hyperinflation	-0.3113	0.4798	0.1860

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Coefficients are event-study estimates relative to the year prior to reform ($T = -1$).

All regressions include country and decade fixed effects, with standard errors clustered at the country level.

Crisis reforms are defined as reforms occurring within one year of inflation exceeding 20 percent and hyperinflation observations (inflation > 50 percent) are excluded.